

PROJECT PLANNING

Project Title: **Script Controlled AC L – Restrict Record Access Based on Field**

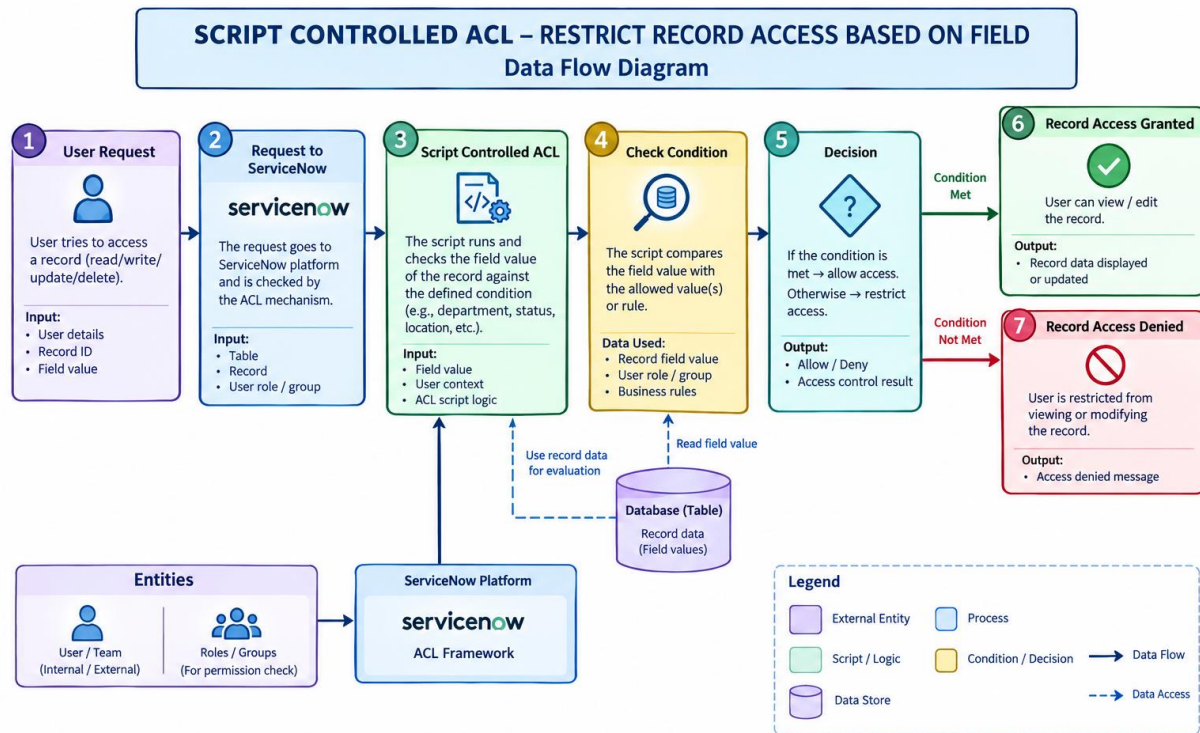
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1. Project Background

ServiceNow is a cloud-based platform used by organizations to manage incidents, requests, users, assets, and other business records. Since these records may contain sensitive information, controlling who can access each record is important. In some situations, normal role-based access control may not be enough. Two records in the same table may require different access levels depending on the value of a particular field. This project introduces a **Script Controlled ACL** to dynamically control record access based on a field value and the user's authorization.

2. Project Planning Approach

The project is planned using the following stages:



Each stage has a specific purpose in developing and validating the access-control solution.

3. Stage 1 – Identify the Security Problem

The first stage is to identify the existing security problem.

Activities

- Study how users currently access ServiceNow records.
- Identify records that require restricted access.
- Identify possible unauthorized access situations.
- Understand why existing access rules may not be sufficient.
- Identify the information that needs protection.
- Determine whether access should depend on a particular field.

Problem Identified

Users may have access to records that they are not authorized to view.

Example

A table may contain:

Record ID	Status
Record 001	Normal
Record 002	Restricted
Record 003	Normal
Record 004	Restricted

The system must distinguish between normal and restricted records.

4. Stage 2 – Understand the User

This stage focuses on understanding the people who use the ServiceNow system.

User Needs

Users need:

- Secure access to records.
- Easy access to permitted information.
- Protection from unauthorized record exposure.
- Clear access rules.
- Consistent security behavior.

User Concerns

Users may ask:

- Who can access this record?
- Why is this record restricted?
- Can unauthorized users view sensitive information?
- Is access automatically controlled?
- What happens when the field value changes?

Expected User Experience

The user should simply open a record. The system should automatically evaluate the access rules without requiring the user to manually request permission for every record.

5. Stage 3 – Requirement Analysis

In this stage, the project requirements are clearly identified.

Functional Requirements

The system should:

1. Identify the required ServiceNow table.
2. Identify the field used for access control.
3. Store appropriate field values.
4. Create an ACL for the required operation.
5. Check the field value using a script.
6. Verify user authorization where required.
7. Allow authorized access.
8. Deny unauthorized access.
9. Apply the rule automatically.
10. Provide consistent access decisions.

Non-Functional Requirements

Security:

Unauthorized users should not access restricted records.

Reliability:

The ACL should work consistently.

Performance:

The access check should execute efficiently.

Maintainability:

The ACL script should be easy to understand and maintain.

Scalability:

The solution should be adaptable if additional access rules are required.

6. Stage 4 – Explore Possible Solutions

Different approaches can be considered before selecting the final solution.

Brainstorming Ideas

1. Role-Based Access Control
2. User-Based Access Control
3. Department-Based Access Control
4. Group-Based Access Control
5. Field-Based Access Control
6. Status-Based Restriction
7. Priority-Based Restriction
8. Multiple Field Validation
9. Dynamic Access Rules
10. Audit and Access Logging

Selected Approach

Field-Based Access Control using Script Controlled ACL

Reason for Selection

The project specifically focuses on restricting record access according to a particular field value. Therefore, an ACL script can be used to evaluate the record dynamically.

7. Stage 5 – Define the Access Rules

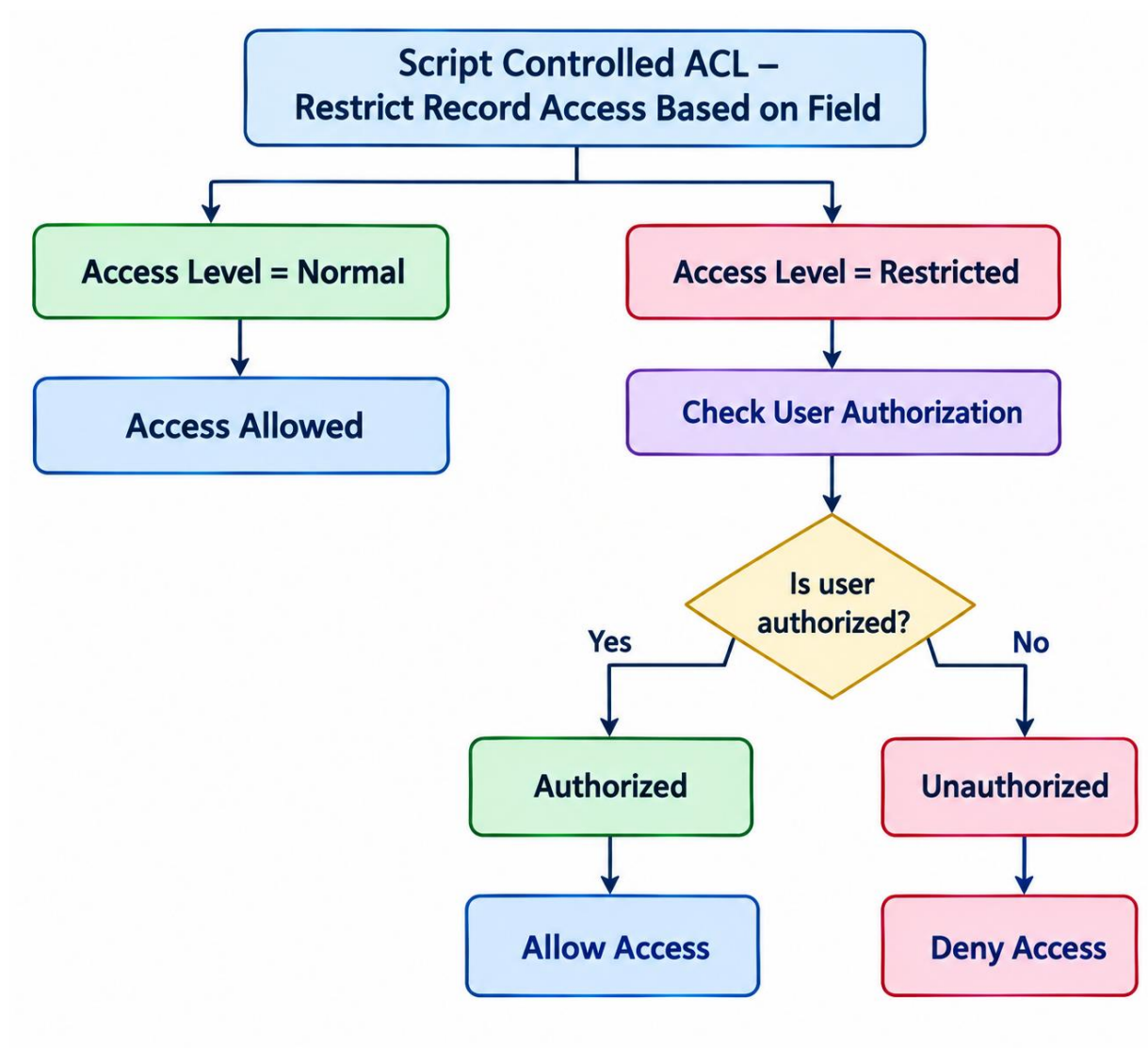
Before implementation, the access rules must be clearly defined

Possible Values

- Normal

- Restricted

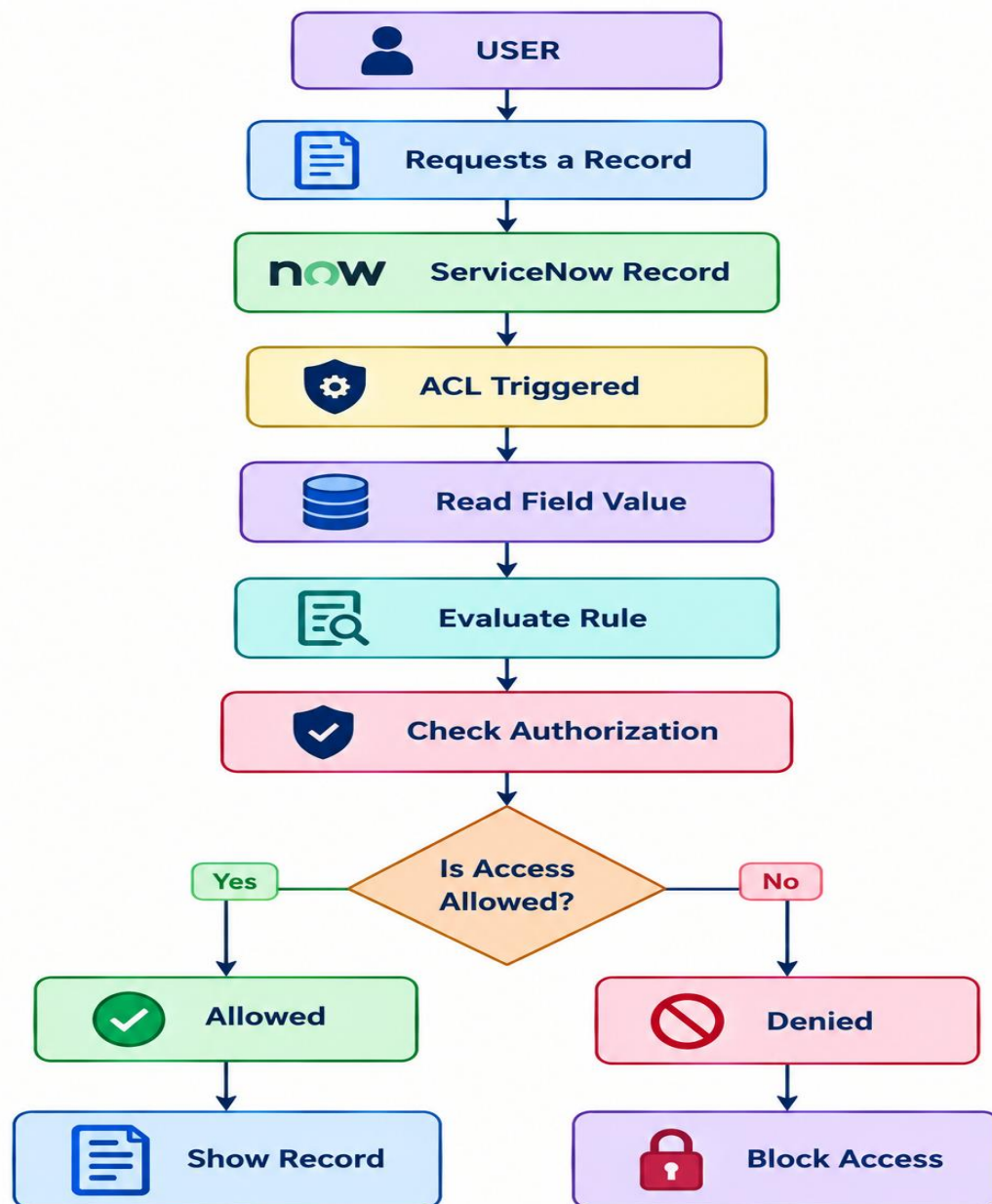
Access Logic



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This makes the security decision clear before the ACL is implemented.

8. Stage 6 – Design the Security Architecture



Main Components

1. **User**
2. **ServiceNow Table**
3. **Access-Control Field**
4. **ACL**
5. **ACL Script**
6. **Authorization Rule**

7. Access Decision

9. Stage 7 – Design the Data and Field Structure

The field used for access control must be identified or created.

Example

FIELD	PURPOSE
Record Number	Identifies the record
Short Description	Describes the record
Access Level	Controls access
Assigned To	Identifies responsible user
State	Shows record status

10. Stage 8 – Implement the ACL

After completing the design, implementation begins.

Implementation Activities

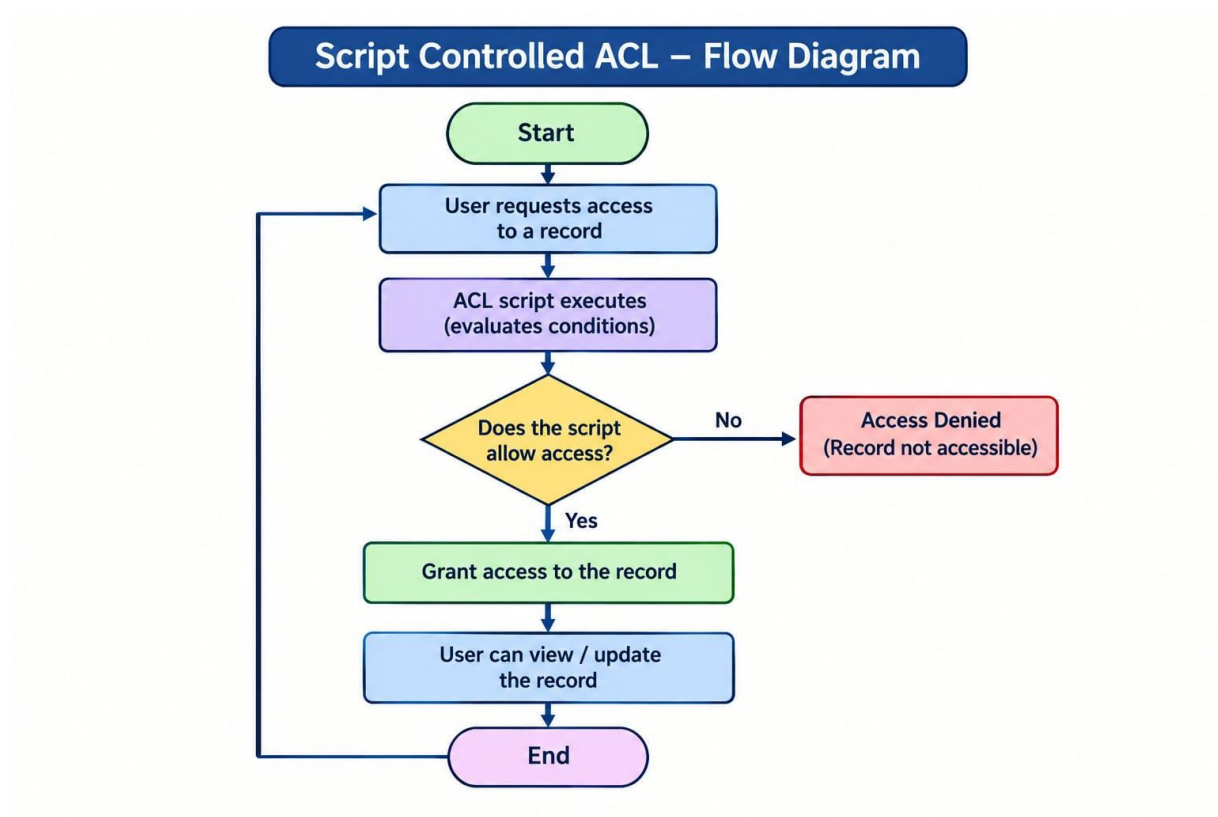
1. Open the required ServiceNow table.
2. Identify or create the access-control field.
3. Create sample records.
4. Assign appropriate field values.

5. Open Access Control configuration.
6. Create the required ACL.
7. Select the table.
8. Select the required operation such as Read.
9. Configure roles/conditions if required.
10. Add the ACL script.
11. Save the ACL.
12. Update the configuration.
13. Verify the ACL behavior.

11. Stage 9 – Script-Controlled Decision

The ACL script is responsible for evaluating the defined conditions.

Logical Model



The script should implement the exact rules defined during the requirement and design stages

12. Stage 10 – Security Challenge

This stage is important because the project should not only show that access works; it should also test situations where access should be blocked.

Security Challenges

Test:

- Authorized user accessing a normal record.
- Authorized user accessing a restricted record.
- Unauthorized user accessing a restricted record.
- Different field values.
- Different roles.
- Different record states.
- Changes to the access-control field.

The purpose is to identify incorrect access decisions.

13. Stage 11 – Testing Plan

Testing should be performed using predefined test cases.

 Testing Evidence			
TEST CASE	FIELD VALUE	USER	EXPECTED RESULT
TC01	Normal	Authorized	Access Allowed
TC02	Restricted	Authorized	Access Allowed
TC03	Restricted	Unauthorized	Access Denied
TC04	Normal	Unauthorized	According to defined rule
TC05	Field Changed	User	Access Re-evaluated
TC06	Different Record	Unauthorized	Access checked again

For each important test:

- Take a screenshot.
- Record the user.
- Record the field value.
- Record the expected result.
- Record the actual result.
- Mark Pass/Fail.

14. Stage 12 – Validation

Validation confirms that the implemented system matches the requirements.

Validation Checklist

- ☐ Correct table selected.
- ☐ Correct field selected.
- ☐ Field values configured.
- ☐ ACL created.
- ☐ Correct operation selected.
- ☐ Required roles configured.
- ☐ Script added.
- ☐ ACL saved and updated.
- ☐ Authorized access tested.
- ☐ Unauthorized access tested.
- ☐ Results documented.
- ☐ Screenshots collected

15. Stage 13 – Documentation

All project activities should be documented.

Documentation Contents

1. Project Title
2. Introduction
3. Background
4. Problem Statement
5. Objectives
6. User Requirements
7. Empathy Map
8. Brainstorming
9. Requirement Analysis
10. Access Rules
11. System Design
12. Workflow
13. Field Configuration
14. ACL Configuration
15. Script Logic
16. Implementation
17. Testing
18. Test Results
19. Screenshots
20. Validation
21. Conclusion

22. Future Enhancements

16. Stage 14 – Final Demonstration

The final demonstration should show the complete working flow.

Demonstration Sequence

Step 1: Explain the problem.

Step 2: Show the selected field.

Step 3: Show sample records.

Step 4: Show the ACL configuration.

Step 5: Explain the script logic.

Step 6: Test an authorized user.

Step 7: Show the allowed result.

Step 8: Test an unauthorized user.

Step 9: Show the denied result.

Step 10: Explain the final outcome.

For a five-member team, the work can be divided as follows:

17. Team Planning

Member 1 – User & Requirement Analyst

- Problem identification
- User requirements
- Empathy map
- Requirement analysis
- Project objectives

Member 2 – System & Field Designer

- Table identification
- Field design
- Access-level design
- Workflow
- System architecture

Member 3 – ACL Implementation

- ACL creation
- Table and operation configuration
- Conditions and roles
- Initial implementation

Member 4 – Script & Security

- ACL script
- Access-control logic
- Security rules
- Security challenge
- Troubleshooting

Member 5 – Testing & Documentation

- Test cases
- Test execution
- Screenshots
- Results
- Final documentation
- Presentation

18. Project Timeline

Phase	Activity	Output
1	Problem Identification	Problem Statement
2	User Understanding	Empathy Map
3	Requirement Analysis	Requirements
4	Brainstorming	Selected Solution
5	Access Rule Definition	Security Rules
6	System Design	Architecture & Flow
7	Implementation	Working ACL
8	Security Challenge	Security Test Results
9	Testing	Test Report
10	Validation	Validation Report
11	Documentation	Final Report
12	Demonstration	Final Presentation

19. Future Enhancements

The project can be extended in the future with:

1. Multiple field-based conditions.
2. Department-based access.
3. Group-based authorization.
4. Role-based restrictions.
5. Dynamic security rules.
6. Access logging.
7. Audit reports.
8. Notifications for denied access attempts.
9. Different access levels such as Normal, Confidential, and Highly Restricted.
10. Centralized security-rule management.

20. Final Project Planning Flow



Final Goal

“To provide secure, dynamic and field-based record access control in ServiceNow using a Script Controlled ACL.”

